

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – STATISTICS I (3.1) – FINAL EXPLANATION: THE MAIZE PRICE MYSTERY AT WAKULIMA MARKET

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Welcome to your Final Explanation! You have spent eight lessons investigating the Maize Price Mystery: a Nairobi family says the price of a 2 kg packet of maize flour at Wakulima Market varies wildly, while their neighbour says it is always about the same. Use the data table below and everything you have learned to answer each section in the space under the prompt. Show every calculation step, write in full sentences, and use words such as data, frequency, mode, mean, median and histogram where they fit. Your answers should always connect back to the driving question: How can we use statistics to tell an honest and useful story about data from our own Kenyan community, and help people make better decisions?

Data: 40 daily prices of a 2 kg packet of maize flour at a Wakulima Market stall over eight weeks.

Price (KSh)	Frequency
98	16
105	12
112	8
120	4

Part 1: From raw chaos to a frequency table

In Lesson 1 the 40 maize prices looked like a chaotic list of numbers that could not settle the argument. (a) Explain why a raw list of data makes it hard to decide whether the family or the neighbour is right. (b) Look at the frequency table above. Check that the frequencies add up to the correct total and say what this total represents. (c) Which price is the mode? What percentage of the 40 days had this price? (d) How many days had a price of KSh 112 or higher? Express this as a percentage of all days.

Write your answer here:

Part 2: Calculating the mean and finding the median

(a) Add a column fx (price $\times$ frequency) to the table and find Σfx . Then calculate the mean price using $\text{mean} = \Sigma fx \div \Sigma f$. (b) Use the frequencies to find the median price. Explain how you decided which prices sit in the middle of the 40 ordered values. (c) Write the mode, median and mean in one line and say whether they are very different from each other.

Write your answer here:

Part 3: What the shape of the data shows

A histogram shows the full shape of the data, not just one number. (a) Look at how the frequencies change as the price goes from KSh 98 up to KSh 120. Describe the shape you would expect a histogram of these prices to have, and name that kind of shape. (b) Explain why this shape helps to explain why the neighbour and the family experienced the market differently. (c) Give one reason why a histogram can tell the story more fully than the mean, mode or median alone.

Write your answer here:

Part 4: Which number tells the truth?

At the school canteen, suppose the daily price of mandazi is about the same each day, except for one day when a sugar shortage in the Rift Valley causes a single very high price of KSh 45. (a) Which of the three measures (mean, median, mode) would be pulled most by that one extreme price, and which would stay stable? Explain why. (b) Now write a short verdict of 3 to 5 sentences answering: 'Who was right in the Maize Price Mystery: the family or the neighbour?' Use at least two of your calculated values from Part 2 and the percentages from Part 1 as evidence.

Write your answer here:

Part 5: Honest statistics for community decisions

A Nairobi family wants to budget for buying one 2 kg packet of maize flour each week at Wakulima Market. (a) Using your results, which price or prices would you advise them to plan around, and which measure(s) did you use? (b) Explain what they should still prepare for. (c) Why is it important to report data and choose a statistic honestly when people use it to make decisions? Give one example from this unit of how statistics could mislead.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Accurate calculation and use of the frequency table (total, mode, percentages, Σfx , mean, median)	All calculations are correct and fully shown, including the fx column, $\Sigma fx = 4204$, mean = KSh 105.10, and the median found from the cumulative frequencies.	Most calculations are correct with steps shown; one small slip, such as a percentage or the median position, does not stop the answer making sense.	Several calculations are missing or incorrect, or the median and mean are confused or no steps are shown.
Understanding of statistical tools and the shape of the data	Clearly explains why raw data cannot settle the argument and why a frequency table and histogram can, and correctly describes the right-skewed shape from the frequencies.	Explains the purpose of tables and histograms and describes the shape, but with limited detail or links to the data.	Gives vague or incorrect statements about tables and graphs, or cannot describe the shape from the frequencies.
Reasoning about mean, median and mode (including the effect of extreme values)	Correctly explains that the mean is pulled by an extreme value while the median and mode stay stable, and explains why each measure suits a different question.	Identifies that the mean is affected by extreme values but the explanation of why is incomplete.	Does not distinguish the measures or gives incorrect reasons.
Evidence-based verdict and community decision-making	Gives a clear verdict that both the family and the neighbour are right, supported by at least two specific values from the data, and offers a sensible budgeting recommendation with a note on honest reporting.	Gives a reasonable verdict and recommendation with some evidence from the data.	Gives an opinion with little or no evidence from the data, or the recommendation does not match the results.

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Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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